Advances in Cognitive Systems 13 (2026) 1-6 Submitted 7/2026; published 9/2026

Integrating Qualitative Models for Reasoning about 3D perspectives and Object Rotations to solve the Purdue Spatial Visualization Test $(Q3D + QOR)_{PSVT}$

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Abstract

This paper explores how qualitative spatial reasoning can be applied to solve analogy challenges such as the Purdue Spatial Visualization Test (PSVT:R) which requires to mentally manipulate 2D representations of 3D objects and find correspondences by rotation. We combine two qualitative models: the qualitative 3D perspective model (Q3D), which represents objects through symbolic consistent depth descriptions from orthographic views, and the qualitative rotation model (QOR), which captures how these descriptors transform under rotations in three-dimensional space. The integration of both models (Q3D+QOR) allow us to compare two objects, by determining whether one can be transformed into the other through a valid sequence of qualitative rotations. Together, the two models provide a symbolic framework for representing spatial structures, reasoning across perspectives, and identifying object correspondence after rotation, while also making explicit the qualitative transformations involved in rotations. This Q3D+QOR model integration has been implemented in Prolog, coupled with an interactive Python solver, and evaluated using the Q3D-PSVT Dataset with 390 multi-choice elements generated across 14 rotation sequences of length one to three. Our results show that the solver recovers a minimal-length rule over $A \rightarrow A'$ and transfers it exactly to object B. Under an exact rule-transfer criterion, the model identifies the designed candidate while rejecting the distractors, showing that Q3D+QOR supports symbolic rotation reasoning without relying on visual similarity, useful prior to agent manipulation tasks.

1. Introduction

Qualitative Spatial and Temporal Representations and Reasoning (QSTR) (Cohn & Renz, 2007; Ligozat, 2011; Sioutis & Wolter, 2021) provides symbolic models for reasoning about spatial and temporal information when precise metric data are unavailable, unnecessary, or cognitively less appropriate. Instead of representing space through numerical coordinates, qualitative models describe relations such as topology, direction, orientation, proximity, intersection and relative position. These representations have been successfully applied across robotics (Mast et al., 2016), computer vision (Suchan, 2022), ambient intelligence (Falomir & Olteţeanu, 2015), architecture and design (Li & Schultz, 2024), geographic information systems (Ali et al., 2017), and education (Santamarta et al., 2023; Kragten et al., 2024), with the most relevant calculi in QSR reported by Dylla et al. (2017).

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Qualitative spatial representations are aligned with what humans often describe when reasoning about space, abstracting away from exact measurements while preserving the spatial relations that are important for the task. This connection with human spatial cognition has motivated the use of qualitative models in sketch recognition (Lovett et al., 2006), when solving cognitive tests about shapes (Pich & Falomir, 2018) or about paper folding and hole punching (Falomir, 2016), when solving analogy problems such as Raven's Progressive Matrices (Lovett & Forbus, 2017), and, closer to the present work, in modeling 2D spatial abstraction during mental rotation tasks (Lovett & Schultheis, 2014).

This paper integrates the Qualitative 3D depth model (Q3D) (Ejarque & Falomir, 2026; Falomir, 2015; Falomir & Oliver, 2016) with the Qualitative Object Rotation (QOR) model (Falomir & Costa, 2025; Falomir, 2026; Falomir et al., 2026) in order to first infer rotations in 3D objects similar to those presented by Shepard-Metzler mental rotation tasks (Shepard & Metzler, 1971) and then apply analogical reasoning to solve the Purdue Spatial Visualization Test (PSVT:R) (Bodner & Guay, 1997; Yoon, 2011).

The contribution of this paper is threefold. First, it shows how Q3D depth descriptions can be transformed under qualitative rotation while remaining in the same symbolic representation. Second, it formulates rotation identification as a finite graph-search problem over complete Q3D states, while allowing object matching through partial visible views. Third, it provides an interpretable output in the form of the shortest qualitative rotation path as a conceptual explanation.

The rest of the paper is organized as follows. Section 2 introduces the Purdue Spatial Visualization Test and reformulates it as a qualitative transformation problem. Section 3 describes preliminaries: the Q3D depth model (Section 3.1) and the QOR rotation model (Section 3.2). Section 4 describes the qualitative rotation operators and the graph-based method for identifying rotated objects from depth matrices. Section 5 reports the results of the rotation generation and sequence recovery experiments. Finally, Section 6 discusses our approach and future work.

2. The problem: the Purdue Spatial Visualization Test

The Purdue Spatial Visualization Test (PSVT) is a classical spatial reasoning task originally composed of different subtests, evaluating different aspects of three-dimensional visualization. These include tasks involving developments, rotations, and views. In this paper, we focus specifically on the rotation component of the test, commonly referred to as the Purdue Spatial Visualization Test: Rotations (PSVT:R) (Bodner & Guay, 1997; Yoon, 2011). In PSVT:R, participants must infer a 3D rotation from one object pair, and apply the same transformation to a second object (Figure 1). In a typical item, the participant is shown an object A and its rotated version A'. A second object B is then presented together with several candidate answers. The goal is to identify which candidate corresponds to object B after undergoing the same rotation that transformed A into A'. The objects used in the task are usually simple 3D block-like structures.

Although the individual components are geometrically simple, the PSVT:R is non-trivial since participants must reason about the global transformation of the object rather than compare local visual features independently. This test requires analogical spatial reasoning: the transformation observed in one object pair must be transferred to a different object. In this sense, the problem

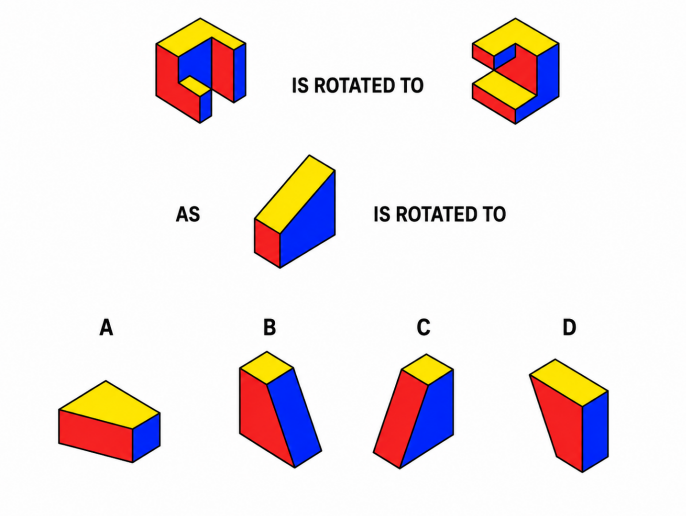


Figure 1. Purdue Spatial Visualization Test: Rotations item.

combines three abilities: representing three-dimensional structures, identifying rotations between two object descriptions, and applying them to a new object.

From the perspective of qualitative spatial reasoning, the PSVT:R task can be reformulated as follows. Given two qualitative descriptions A and A' of a first object before and after rotation, we ask whether there exists a sequence of qualitative rotations that transforms A into A': $A' = \rho_n(\cdots \rho_1(A))$. The inferred rotation sequence $R = \langle \rho_1, \ldots, \rho_n \rangle$ is then applied to a second object B. The correct answer is the candidate C_i whose qualitative description matches the transformed version of B: given a set of candidates $\mathscr{C} = C_1, \ldots, C_m$, the task is to find $C^* = C_i \in \mathscr{C} \mid C_i = R(B)$, where each ρ_i denotes a qualitative rotation operator. Thus, the central problem is not only to decide whether two objects are equivalent, but to recover a transformation from one object pair and reuse it on a different object.

In the following sections, we use the PSVT:R as a test case for showing how qualitative descriptors of depth and rotations combined (Q3D+QOR) can solve analogical rotation challenges and provide conceptual explanations. The goal is not to simulate continuous mental imagery, but to show that symbolic qualitative models can represent the relevant spatial structure, infer a transformation between object states, transfer that transformation to a new object, and identify the correct candidate through an interpretable rotation path.

3. Preliminaries: Q3D and QOR

3.1 The Qualitative 3D Depth model (Q3D)

When reasoning about real objects in space, humans think in terms of three-dimensional volumes and multiple perspectives. The Q3D model (Falomir, 2015; Falomir & Oliver, 2016) formalizes this

intuition by encoding orthographic projections using symbolic depth values and relations between these depth values at neighboring object boundaries: $Q3D = \langle Q3D_{dRS}, Q3D_{rRS} \rangle$.

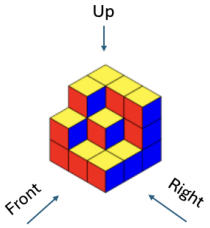
The Q3D Depth descriptor ($Q3D_d$) of an object is defined by three views:

$$Q3D_{RS} = \{F, R, U \in View \mid Q3Dmatrix \subseteq N_{depths}\}, \text{ such that}$$

$$\begin{aligned} &view(F, Object, H, W, Q3Dmatrix(H, W)),\\ &view(R, Object, H, L, Q3Dmatrix(H, L)),\\ &view(U, Object, L, W, Q3Dmatrix(L, W)). \end{aligned}$$

where F, R, and U denote the *Front*, *Right*, and *Up* perspectives, while H, W, and L represent the height (H), width (W), and length (L) of the object's bounding box, $N_{depths} = \{a, b, c, d, \cdots, *\}$ and each matrix entry belongs to the symbolic set: $Q3Dmatrix(D_1, D_2) \subseteq \{a, b, c, d, \ldots, *\}$, where D_1 and D_2 are the dimensions of the matrix (pairs of H, W, L) and the elements of each matrix describe ordinal depth layers composed of equally sized volumetric units: a denotes the surface of the object, b the first level of depth, c the second level, d the third level, and so on. The symbol $*$ indicates that all volumes along a depth column have been removed, so there is a hole. These symbols encode relative ordering rather than metric distance.

Figure 2 presents an object described by the Q3D descriptor for a $3 \times 3 \times 3$ grid, showing the *Front*, *Right*, and *Up* matrices and their corresponding logic facts. Note that a smaller volume size could equally have been used, producing larger matrix dimensions for the same descriptor.



Q3D descriptor for this 3x3x3 object:

Front	*Right*	*Up*
$[b,c,c]$	$[*,c,a]$	$[a,a,a]$
$[a,b,c]$	$[c,b,a]$	$[a,b,c]$
$[a,a,a]$	$[a,a,a]$	$[b,c,c]$

Logically as Prolog facts:

$view(front, obj, 3, 3, [[b,c,c],[a,b,c],[a,a,a]]).$
$view(right, obj, 3, 3, [[*,c,a],[c,b,a],[a,a,a]]).$
$view(up, obj, 3, 3, [[a,a,a],[a,b,c],[b,c,c]]).$

Figure 2. Example of (a) an object observed from *Front*, *Right*, and *Up* sides; (b) an intuitive representation of the Q3D descriptor; (c) the logic facts produced by the Q3D descriptor for this specific object.

Beyond representation, Q3D imposes geometric compatibility constraints between views: a description is *consistent* (Falomir, 2015) if every pair of views (*Front–Right*, *Front–Up*, *Right–Up*) can be jointly realised as orthographic projections of a single three-dimensional object. Intuitively, two adjacent views are consistent if there exists a volumetric arrangement whose projections simultaneously satisfy the qualitative depth relations encoded in both, since the values along the shared edge between two views must agree. The full deterministic constraints for each pair of adjacent views are given in Falomir (2015).

The Q3D reasoning model ($Q3D_r$) provides deterministic rules for inferring depth values on the occluded sides of an object (*Back*, *Left*, *Down*) by exploiting space continuity and depth interrelations between opposite faces: for instance, a maximal depth value on *Front* entails a surface value

(a) on the opposite *Back* face, with analogous rules relating *Right*/*Left* and *Up*/*Down*, and holes ($*$) propagating from a view to its opposite side.

The $Q3D_r$ depth resoning model introduces: (i) a symbolic volumetric representation, (ii) cross-view consistency constraints, and (iii) inference mechanisms based on space continuity and depth interrelations between opposite sides. It shows that spatial structures and depth relations can be inferred from complementing perspectives. The Q3D reasoning model was first implemented and tested in a SWI-Prolog application using 3×3×3 objects (Falomir, 2015), then generalized in Ejarque & Falomir (2026), and also applied in a videogame (Falomir & Oliver, 2016) aimed at training players' spatial reasoning skills. Ejarque & Falomir (2026) showed that the Q3D model allows any dimension and it affords inference rules for discovering volumes in occluded sides.

The Q3D model represents an object through symbolic depth matrices associated with orthographic views such as *Front*, *Right*, and *Up*. In this paper, we use complete six-face Q3D states, including *Front*, *Back*, *Left*, *Right*, *Up*, and *Down*, so that rotations can be applied consistently even when some faces are not visible in the test item.

3.2 The Qualitative Object Rotations model (QOR)

The Qualitative Model of Object Rotation (QOR) by Falomir et al. (2026) formalizes how an object's perspectives evolve during rotation manipulation, combining a Qualitative Object Descriptor (QOD) (Falomir & Costa, 2025) with a Rotation Reference System (QOR_{RS}) that defines the rotation movements applicable to a QOD: $QOR = \langle QOD_{RS}, QOR_{RS} \rangle$.

The Qualitative Object Descriptor (QOD) embeds an object O within a conceptual bounding box with 6 canonical sides parallel in pairs. A view of the object, $view_{O_{xyz}}$, is defined by three mutually perpendicular sides (x, y, z), each characterized by a *feature* (content), a *location*, and an *orientation*:

$$side(Scn, Obj, Location_{RS}, Feature_{RS}, Orientation_{RS})$$

where $Location_{RS}$ identifies each side with one of $\{F, R, U, D, B, L\}$ (*Front*, *Right*, *Up*, *Down*, *Back*, *Left*), together with the corresponding bounding-box plane; $Orientation_{RS}$ captures the absolute or relative turning of a side content in 90° increments (e.g., +q represents *turned a quarter clockwise*); and $Feature_{RS}$ captures the content of a side, ranging from raw sensory data (pixels, voxels) to the symbolic concept assigned to it through an external abstraction process (manual annotation or automated perception). In this paper, the features describing our objects are volumes described by the $Q3D_{RS}$ descriptor: $Feature_{RS} = \langle Q3D_{RS} \rangle$. This is the key integration point between the two models: each side of the QOD bounding box carries a Q3D depth matrix as its feature.

The Qualitative Object Rotation (QOR) model relates rotation labels and their geometric definitions through the relation $rotation(Rotation_N, Rotation_G, Axis)$, where the axes are defined by the lines through the centroids of opposite object sides, following the standard Euler convention relative to the object's centroid; $Rotation_G$ specifies the rotation angle in discrete 90° increments, and $Rotation_N$ assigns it a qualitative label such as *towards-up* or *towards-up-right*. Table 1 lists the QOR rotations used in this paper; their effect on face location and orientation is detailed in (Falomir et al., 2026; Falomir, 2026) and summarized next in Table 2, and applied directly to Q3D states in Section 4.

Table 1. QOR_{RS} specification using icon representation of rotations to increase the readability.

Icon	*rotation*(Rotation$_N$, Rotation$_G$, Axis).
$\uparrow$	*rotation*(towards-up, 270, x).
$\downarrow$	*rotation*(towards-down, 90, x).
$\leftarrow$	*rotation*(towards-left, 270, y).
$\rightarrow$	*rotation*(towards-right, 90, y).
$\curvearrowright$	*rotation*(towards-up-right, 270, z).
$\curvearrowleft$	*rotation*(towards-up-left, 90, z).

Table 2. Inference table which relates the initial location and the final location of a feature with the corresponding rotation movement and the final relative orientation.

from/to	**front**	**back**	**up**	**down**	**right**	**left**
front	$\curvearrowright^{+q}$ ∨ $\curvearrowleft^{-q}$		$\uparrow$	$\downarrow$	$\rightarrow$	$\leftarrow$
up	$\downarrow$	$\uparrow^{2q}$	$\rightarrow^{-q}$ ∨ $\leftarrow^{+q}$		$\curvearrowright^{+q}$	$\curvearrowleft^{-q}$
down	$\uparrow$	$\downarrow^{2q}$		$\rightarrow^{+q}$ ∨ $\leftarrow^{-q}$	$\curvearrowleft^{-q}$	$\curvearrowright^{+q}$
right	$\leftarrow$	$\rightarrow$	$\curvearrowleft^{-q}$	$\curvearrowright^{+q}$	$\uparrow^{+q}$ ∨ $\downarrow^{-q}$	
left	$\rightarrow$	$\leftarrow$	$\curvearrowright^{+q}$	$\curvearrowleft^{+q}$		$\uparrow^{-q}$ ∨ $\downarrow^{+q}$
back		$\curvearrowleft^{+q}$ ∨ $\curvearrowright^{-q}$	$\downarrow^{2q}$	$\uparrow^{2q}$	$\leftarrow$	$\rightarrow$

Location and Orientation change. By observing the relation between a rotation action and: (i) the changing location and (ii) the relative orientation change, the inference Table 2 can be built, where the leftmost column presents the initial feature location, while the top row presents the final location. The intersection cell specifies the rotation (Rotation$_N$) required to perform this transformation according to QOR_{RS} specification (Table 1). The superscript ΔO in $\text{Rotation}_N^{\Delta O}$ specifies the relative orientation increase or decrease applied to the feature as the result of the rotation. The absence of a superscript implies that the orientation of the feature remains the same. Empty cells indicate that QOR rotation cannot directly move a feature from the starting location to the ending location (e.g. between opposite sides).

4. Method: Mixing Q3D+QOR for Rotation Inference and Face Transformation

The previous section outlined the Q3D or QOR formalisms. This section presents our new contribution, that is, how the Q3D and QOR models are composed and searched over. For that, we consider the notion of *view* in the Q3D model equivalent to *side* in the QOD model, which will be referred as object *side* from now on.

4.1 Problem Representation

We represent each object as a discrete 3D structure with six oriented faces: *Front*, *Back*, *Left*, *Right*, *Up*, and *Down*, as both Q3D and QOR models propose. Each object *side* is encoded as a depth

matrix, $Feature_{RS} = \langle Q3D_{RS} \rangle$, so each object preserves both its global spatial structure and the local depth feature on each *side*.

This representation is appropriate since a rotation acts as two levels. First, it permutes which *side* occupies each spatial position in the object. Second, it may reorient the depth matrix associated with a face, even when the *side* remains in the same relative location. As a result, the state of the object after rotation is determined by both *side* relocation and in-plane matrix transformation. This separation is the key abstraction used throughout the method: the object is treated as a structured set of depth matrices, and every transformation is defined in terms of how those matrices move and how they are rotated internally.

4.2 Applying Rotations to Q3D Model

A rotation is represented as a transformation of a complete six-face state. In our implementation, each rotation is specified by a declarative rule indicating three pieces of information: the source face, the transformation that must be applied, and the target face. A rotation is therefore not encoded as a numerical operation over 3D coordinates, but as a symbolic transformation over Q3D depth maps. The possible transformations of a face matrix are:

$$none, \quad rot_{90}^{cw}, \quad rot_{90}^{ccw}, \quad rot_{180}.$$

These transformations only change the position of the symbols inside the matrix; they do not change the symbols themselves. Therefore, a cell labeled a, b, c, or $*$ preserves its qualitative depth meaning after the rotation.

The QOR rotation model (Falomir, 2026) distinguishes between rotations around different object axes (Table 1). Pitch rotations or rotations in x axis (*towards_up*, *towards_down*) affect the vertical arrangement of the object; for a rotation *towards_up*, *sides* are reassigned as

$$F \to U, \qquad U \to B, \qquad B \to D, \qquad D \to F.$$

Yaw rotations or rotations in y axis (*towards_left*, *towards_right*) affect the horizontal arrangement; for *towards_left*, the side faces are reassigned as

$$R \to F, \qquad F \to L, \qquad L \to B, \qquad B \to R.$$

The implementation also considers Roll (in-plane) rotations or rotations in z axis, corresponding to 90-degree turns around the observer's viewing axis (*towards-up-right* and *towards-up-left*). Here, the *Front* and *Back* faces remain in place while their features (3D matrices) change orientation and the neighbouring *sides* cycle around them; for *towards-up-right*, $U \to R \to D \to L \to U$, with the inverse cycle for *towards-up-left*. This allows the model to capture rotations that do not change *side* location, but do change the orientation of the object within the image plane.

This preserves the qualitative character of the representation: the method does not reconstruct a metric 3D model and rotate it geometrically. Instead, it reasons directly over Q3D descriptions, relocating symbolic depth maps and reindexing their matrix positions. As a result, the transformed object remains expressed in the same qualitative language as the original, allowing rotated states to be compared directly with other Q3D descriptions.

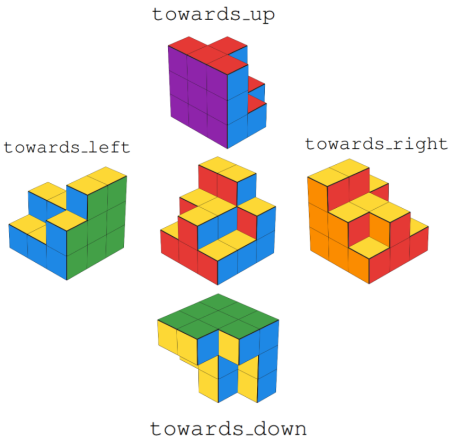


Figure 3. Schematic effect of viewpoint-changing after rotations in axis x and y on a Q3D object.

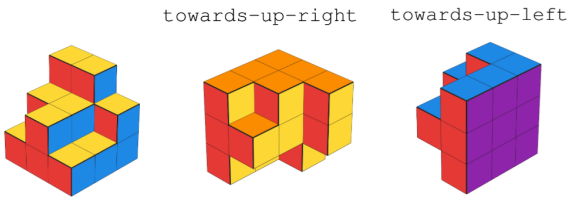


Figure 4. Schematic effect of Roll (in-plane) rotation in z axis on a Q3D object.

4.3 Rotation Identification from Depth Matrices

Once rotations have been applied to Q3D states, the next challenge consists on identifying whether two qualitative descriptions correspond to the same object under rotation. We formulate this as a search over a finite graph which represent *side* location-orientation change after a 90-degree rotation which results from expanding Table 2 (Falomir et al., 2026; Falomir, 2026). Each node is a complete six-*side* Q3D state, and each directed edge corresponds to one elementary rotation: *towards_up*, *towards_down*, *towards_left*, *towards_right*, *towards-up-right*, and *towards-up-left*.

The graph is generated from the initial object by repeatedly applying all available rotations to every newly discovered state. Whenever a new six-face configuration is produced, it is added as a node; if the same configuration has already been reached through another sequence, it is not expanded again. This produces the finite set of reachable orientations for the object, together with the rotation labels that connect them.

The order and size of this graph are bounded by the number of finite orientations in an object (which increase or decrease a quarter depending on the 90-degree rotation performed) (Falomir et al., 2026; Falomir, 2026). A complete orientation can be specified by choosing which of the six *sides* is in the *Front* location and, once this is fixed, which of the four remaining *sides* is in located *Up*. This means that an asymmetric object has at most $6 \cdot 4 = 24$ distinct orientation states, so the

graph order is $|V| \leqslant 24$. Note that, as the implementation applies the six elementary rotation operators from each discovered node, the resulting graph has six directed labelled outgoing transitions per node; therefore $|E| = 6|V|$, and in the generic asymmetric case $|E| = 6 \cdot 24 = 144$.

The comparison between objects can then be performed at two levels. The rotation graph is constructed using the complete six-*side* state, because the hidden *sides* are needed to apply rotations consistently. However, the matching step is restricted to the visible perspectives used in the test item: in our case, *Front*, *Right*, and *Up*, abbreviated as FRU. Two states are considered observationally equivalent when their FRU signatures coincide ($FRU(S_i) = FRU(S_j)$).

This distinction is important for modeling mental rotation tests, since a participant is normally shown only a limited view of an object, not its complete six-face Q3D description. Thus, our method uses the full state internally to guarantee coherent rotations, but identifies matches through the visible depth matrices: given a fixed object and a transformed candidate, it retrieves all graph nodes sharing the fixed object's FRU description and all nodes sharing the candidate's.

The inferred transformation is then obtained by searching for a shortest path between these two sets of nodes. Since every edge corresponds to one rotation, the shortest path is the sequence with the minimum number of qualitative rotations connecting the two observed descriptions:

$$S_i \xrightarrow{\rho_1,\ldots,\rho_n} S_j, \qquad FRU(S_i) = FRU(O_1), \quad FRU(S_j) = FRU(O_2).$$

If path of rotations exists, the candidate is classified as a rotated version of the fixed object. Note that, the sequence $\rho_1, \ldots, \rho_n$ is not only a computational result, but also an interpretable description of how the first object can be rotated to obtain the the second one, as illustrated by the example in next Section 5.

5. Evaluation

We evaluate the method on the Purdue Spatial Visualization Test (PSVT) analogy task. Each question presents Object A, its rotated version A', and Object B, and asks which of four candidates is Object B transformed by the same hidden rotation sequence that maps A to A'. The evaluation focuses on three aspects: whether rotations generate coherent Q3D states that can serve as the hidden rule, whether the inferred rule can be reliably applied to a new object, and how the solver handles distractor candidates that are reachable from B by some rotation but do not match the $A \rightarrow A'$ rule.

5.1 Forward Rotation Generation

The first result concerns the generation of rotated Q3D descriptions that define the hidden PSVT rule. Given Object A as a six-*side* Q3D state, each rotation operator in the vocabulary $\{$*towards_up*, *towards_down*, *towards_left*, *towards_right*, *towards-up-right*, *towards-up-left*$\}$ produces A' in the same qualitative format. That is, after each rotation, the object is still represented by the six canonical perspectives *Front*, *Back*, *Left*, *Right*, *Up*, and *Down*, and each perspective is again encoded as a Q3D depth matrix.

This shows that the rotation model is closed over the Q3D representation. This is important because it allows rotations to be composed, meaning that a qualitative rotation produces another valid Q3D state that can immediately serve as input to the next rotation. The PSVT dataset uses 14

rotation sequences of length one, two, and three, covering all six elementary operators and selected two-step and three-step combinations. A representative two-step sequence is:

$$A \xrightarrow{towards_left} A_1 \xrightarrow{towards_up} A'.$$

Each intermediate state remains directly comparable with other Q3D descriptions, so the same rule can be extracted symbolically and re-applied to Object B. Figure 5 illustrates this process for one

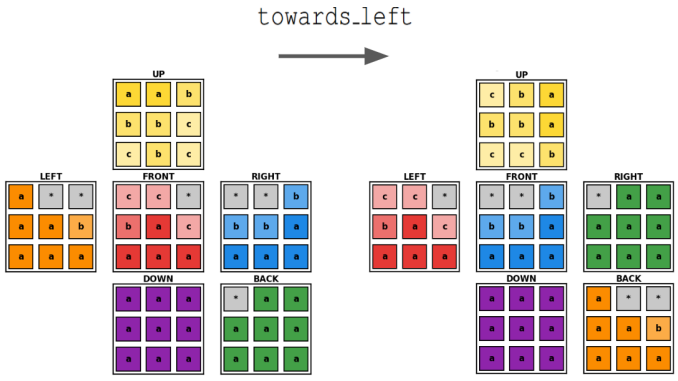


Figure 5. Example of a Q3D object before and after applying a qualitative rotation. Each state is represented by the six canonical depth matrices, showing that the rotation operators generate new Q3D states in the same symbolic format.

example object: the image on the left shows the original Q3D state, while the image on the right shows the state obtained after applying one qualitative rotation from QOR. The symbols inside the matrices are preserved, but their positions and *side* assignments change according to the selected rotation. Rotations around the vertical axis y reassign the lateral *sides* while rotating the *Up* and *Down* matrices in place; rotations in axis z or in-plane keep the *Front* and *Back sides* in their locations while rotating their orientations and cycling the surrounding side location. This side-by-side transparency is useful for verifying that the operators behave as expected before applying them as the rule in the PSVT task.

5.2 Rule Transfer to Object B

The core PSVT inference step is to apply the hidden $A \rightarrow A'$ sequence to Object B and compare the result against each candidate. The solver proceeds as follows. First, it extracts the rotation sequence $\rho_1, \ldots, \rho_n$ from the $A \rightarrow A'$ transformation by searching the rotation graph of Object A for the shortest path between the FRU signature of A and the FRU signature of A'. Second, it applies the same sequence directly to Object B to obtain the expected result B':

$$B \xrightarrow{\rho_1} B_1 \xrightarrow{\rho_2} \cdots \xrightarrow{\rho_n} B'.$$

Third, each candidate is compared against B' by exact equality of its six-side Q3D state. The candidate whose state matches B' is selected as the answer.

This design is deliberate: the solver does not choose the candidate nearest to B in the rotation graph, because wrong-rotation distractors can also be reachable from B by some valid sequence. By anchoring on the exact $A \rightarrow A'$ rule applied to B, the solver avoids selecting a reachable but rule-inconsistent candidate.

The recovered rule is not necessarily unique. A PSVT:R question is generated by applying a specific sequence, but an equivalent orientation of A' may be reachable through shorter or different sequences. The solver returns the shortest path in the rotation graph, which corresponds to the most suitable qualitative explanation of the demonstrated transformation. For example, if the generating sequence is *towards_left x 4 times* followed by *towards-up-right*, the four horizontal rotations cancel out and the recovered rule is simply:

$$\rho_{\text{inf}} = \textit{towards-up-right}.$$

The same minimal rule is then applied to B, producing the expected candidate.

5.3 Results

Base objects. The dataset is build from six base $3 \times 3 \times 3$ objects, each defined as a complete six-sides Q3D state (*Front*, *Back*, *Left*, *Right*, *Up*, *Down*) extracted from Ejarque & Falomir (2026); Falomir (2015); Falomir & Oliver (2016). Figure 6 shows all six templates as reconstructed from these depth views. They were chosen to span a range of structural complexity, measured as the number of unit cubes occupied out of the 27 available in the bounding box: from sparse configurations (*object5*, *object6*: 11/27) to dense ones (*object3*, *object4*: 22/27), with *object1* and *object2* in between.

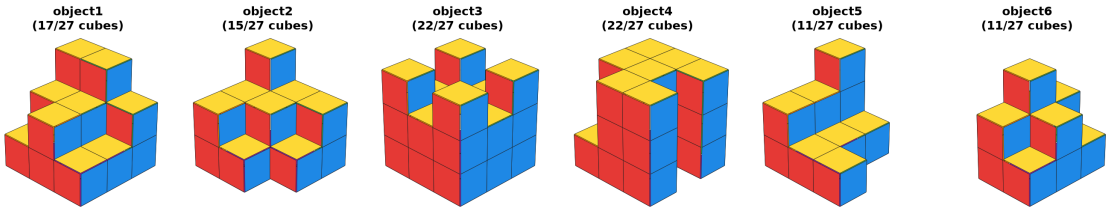


Figure 6. The six base template objects used to build the PSVT analogy dataset, with the number of occupied unit cubes (out of 27) shown under each name.

Predefined rotation sequences. Every item is generated from one of the 14 predefined rotation sequences, built from the six elementary QOR rotations in Table 1: the 6 single-rotation sequences, 6 two-rotation sequences pairing a directional rotation with an in-plane rotation (e.g. *towards_up*→*towards_up_right*, *towards_up_right*→*towards_left*), and 2 three-rotation sequences (*towards_up* → *towards_left* → *towards_up_right* and its mirror *towards_down* → *towards_right* → *towards_up_left*). This fixed list is curated rather than sampled from all $6 + 6^2 + 6^3 = 258$ possible combinations, to keep the benchmark small while covering easy (length 1), medium (length 2), and harder (length 3) items. For every ordered pair of distinct base objects and every sequence,

a sample is kept only if the sequence produces a visible change on *both* A and B (i.e. the resulting Q3D signature differs from the original), discarding accidental no-op rotations caused by object symmetries.

Dataset description. The resulting dataset contains 390 items (four candidates each, 1170 distractor options in total). Sequence length is not uniform by design: 160 items use a length-1 sequence, 180 use length-2, and 50 use length-3, reflecting that only 2 of the 14 predefined sequences reach length 3. All 6 base objects are used both as A and as B, each participating in 55–67 items; the 30 distinct ordered (A,B) object pairs are each covered by 11–14 items across the 14 rotation sequences.

Evaluation protocol. Correctness is judged by exact match, not manual or visual inspection. Each object state (i.e. an item's A, A', B, and its four candidates) is represented as a six-*side* Q3D state. Given the demonstrated rule (the rotation sequence used to turn A into A'), the solver applies the same sequence to B to obtain the expected result B', and compares each of the four candidates against B' by exact signature equality. A sample counts as solved, if and only if, exactly one candidate's signature equals that of B' and it is the candidate designated as correct at generation time.

Example. Each item in the PSVT:R analogy dataset reported here is solvable by construction: the correct candidate is always the direct image of B under the demonstrated rule, and the remaining three options are distractors built to be visually plausible but rule-inconsistent. An example of a distractor can be seen in Figure 7.

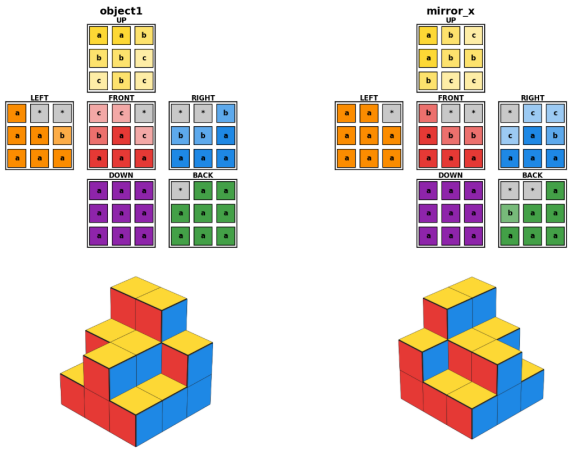


Figure 7. Representative distractor option, generated by mirroring one of the base objects (object1).

The relevant question is therefore not how many of a solvable/unsolvable split were answered correctly, but (i) whether the exact-match criterion picks out the designated correct candidate for every item, and (ii) whether any distractor satisfies that same criterion, which would indicate a false positive. Each item is solved correctly under the exact-match criterion, and no distractor option anywhere in the dataset ever satisfies the rule-equality check. This is expected given the deterministic, symbolic nature of the Q3D+QOR pipeline. We emphasize that this result should be read as a verification of implementation correctness. Figure 8 shows a sample from the dataset,

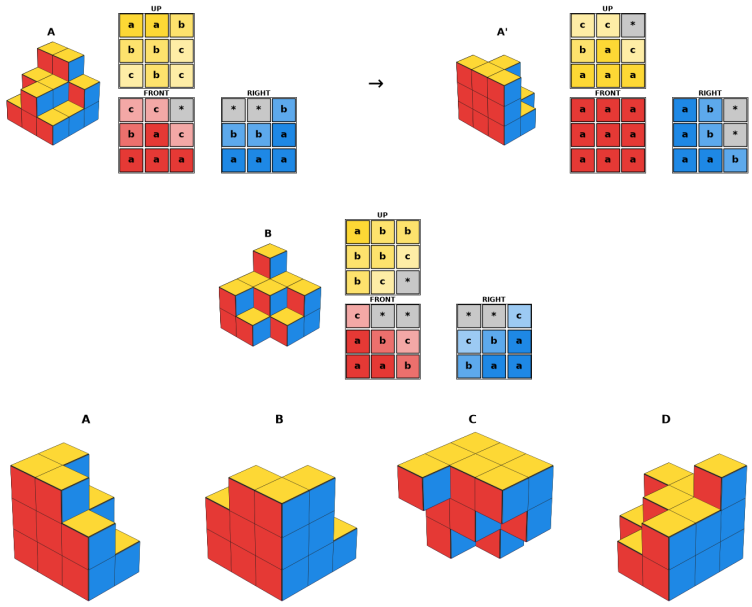


Figure 8. Worked example from the dataset. Correct candidate is **A**.

with the rotation rule *towards_up*. The solver applies this sequence to Object B and computes B'. Then, it compares each candidate against B' by exact Q3D equality, returning the one that matches. The reasoning trace produced by the solver lists, for each candidate, whether it matches the rule, whether it is reachable from B by any rotation path, and the length of the shortest path. This makes the decision transparent: the selected answer is not merely the closest reachable orientation but the unique orientation of B that the demonstrated rule predicts.

6. Discussion and Future Work

This paper presented a qualitative approach for reasoning about object rotations in order to solve the Purdue Spatial Visualization Test about Rotations (PSVT:R). By combining Q3D depth descriptors with Qualitative Rotation operators (Q3D+QOR), our approach can represent objects as symbolic six-side states, generate rotated descriptions, and recover short transformation paths from visible *Front*, *Right*, and *Up* views. We evaluated our $(Q3D + QOR)_{PSVT}$ approach creating the 390-item Q3D-PSVT dataset. Our results confirms that this symbolic pipeline is internally consistent: the solver recovers a minimal-length rule from $A \rightarrow A'$, transfers it exactly to B, and rejects every distractor, without ever comparing pixel or mesh-level appearance between candidates. We see this particularly relevant as a preliminary step prior to agent manipulation tasks, where an agent must first establish that an observed object and a target configuration are related by a well-defined rotation before committing to a physical re-orientation action. A symbolic equivalence check of this kind could act as a lower-cost gate ahead of a full geometric motion-planning stage.

As future work, we intend to extend our $(Q3D + QOR)_{PSVT}$ approach beyond controlled block objects. One direction is to connect Q3D descriptions with perceptual input, such as RGB-D images voxel grids built from point clouds, so that the model can reason about real objects under partial observation. This would also support affordance checking, where qualitative features such as holes, flat regions, or accessible surfaces determine whether an object can be grasped, stacked, inserted, or used as support.

A second direction concerns grounding the approach closer to how the PSVT:R is actually used. Beyond its paper-and-pencil form, spatial visualization is sometimes assessed or trained with physical solid models that participants manipulate by hand. An interesting application of our pipeline would be to automatically derive Q3D descriptions from such physical replicas (e.g. via depth cameras) in order to validate item answer keys, generate new items, or check consistency between a physical prototype and its intended rotation rule. This would tie naturally into robotics applications built on the same representation. Since Q3D makes both visible and occluded structure explicit, an agent could use the rotation graph as a compact space of candidate re-orientations for Best-Next-View planning, choosing the next viewpoint expected to resolve the most ambiguity about an object's identity or pose. The same graph could also serve as a symbolic motion-primitive space for object re-orientation: given an observed pose and a desired target pose (possibly only partially visible), the agent could first search for a qualitative rotation path between them, exactly as done here for $A \rightarrow A'$, and use the recovered sequence to guide or validate a subsequent low-level manipulation policy.

As a final direction, the implementation should be generalized to arbitrary $H \times W \times L$, just like Q3D, and evaluated on larger object sets, symmetric objects, and partially observed cases. In particular, the dataset reported here draws from a deliberately small, curated set of 14 rotation sequences rather than the full combinatorial space of length-1-to-3 compositions of the six elementary operators ($6 + 6^2 + 6^3 = 258$). Scaling to that full space, and to genuinely symmetric objects where several rotation sequences are qualitatively indistinguishable, is expected to make shortest-path rule recovery ambiguous in a way the current dataset does not exercise, and would test the solver under conditions closer to real PSVT:R difficulty.

Code Availability

The code and dataset used in this work are available here.

Acknowledgments

We acknowledge the funding by the Wallenberg AI, Autonomous Systems and Software Program (WASP) awarded by the Knut and Alice Wallenberg Foundation, Sweden.

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